Graph-Based Cyber Intrusion Detection System

Abstract

With the rapid growth of networked systems, cyber attacks have become increasingly sophisticated, making traditional signature-based intrusion detection systems ineffective against unknown threats. To address this limitation, this project proposes a graph-based cyber intrusion detection system leveraging machine learning and graph analytics. In this approach, network traffic is modeled as a graph where nodes represent entities such as IP addresses and ports, and edges represent communication between them. Structural features including node degree, clustering coefficient, and neighborhood relationships are extracted to capture complex interaction patterns. The system is further enhanced by incorporating Graph Neural Network (GNN)-based concepts to improve the learning of hidden relationships within graph-structured data.

The model is trained and evaluated on the UNSW-NB15 dataset, incorporating data balancing techniques to address class imbalance. Performance is assessed using advanced metrics such as Precision, Recall, and F1-score, and compared with traditional machine learning models to demonstrate improved detection accuracy, particularly for minority attack classes. Additionally, a real-time monitoring interface is developed using Streamlit to enable live prediction and visualization of network threats. The proposed system effectively identifies hidden patterns and anomalies, providing a more scalable, accurate, and robust solutions compared to traditional intrusion detection approaches.

Keywords:
Cyber Security, Intrusion Detection, Graph Analytics, Machine Learning, Graph Neural Networks, Network Traffic

Ms. M. Sri Sevitha
Name of the Guide

O. Ranitha
25WH1D5802